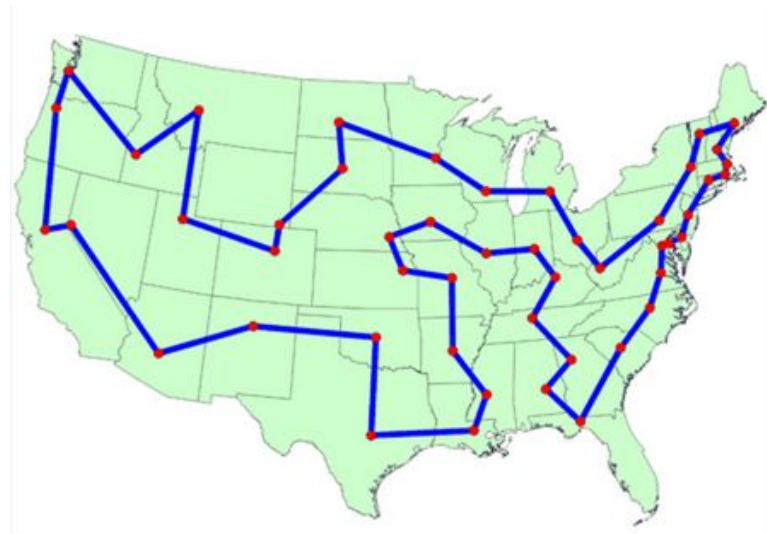




Lecture 11

Introduction to Asymptotic Analysis

CS61B, Fall 2025 @ UC Berkeley

Josh Hug and Peyrin Kao

Sp26 TODO: Consider starting with big Theta, big O, big Omega as purely mathematical, then move on to runtime analysis later.

Goal: Measuring Code Efficiency

Lecture 11, CS61B, Fall 2025

Goal: Measuring Code Efficiency

Intuitive Runtime Characterizations

- Clock Time
- Exact Operation Counting
- Exact Count Exercise

Asymptotic Analysis

- Why Scaling Matters
- Computing Worst Case Order of Growth (Tedious Approach)
- Computing Worst Case Order of Growth (Simplified Approach)

Asymptotic Notation

- Big Theta (a.k.a. Order of Growth)
- Big O

An engineer will do for a dime what any fool will do for a dollar.

Efficiency comes in two flavors:

- Programming cost (course to date. Will also revisit later).
 - How long does it take to develop your programs?
 - How easy is it to read, modify, and maintain your code?
 - More important than you might think!
 - Majority of cost is in maintenance, not development!
- Execution cost (from today until end of course).
 - How much time does your program take to execute?
 - How much memory does your program require?

Example of Algorithm Cost

Objective: Determine if a sorted array contains any duplicates.

- Given sorted array A , are there indices i and j where $A[i] = A[j]$?

-3	-1	2	4	4	8	10	12
----	----	---	---	---	---	----	----

Silly algorithm: Consider every possible pair, returning true if any match.

- Are $(-3, -1)$ the same? Are $(-3, 2)$ the same? ...

Better algorithm?

Example of Algorithm Cost

Objective: Determine if a sorted array contains any duplicates.

- Given sorted array A , are there indices i and j where $A[i] = A[j]$?

-3	-1	2	4	4	8	10	12
----	----	---	---	---	---	----	----

Silly algorithm: Consider every possible pair, returning true if any match.

- Are $(-3, -1)$ the same? Are $(-3, 2)$ the same? ...

Today's goal: Introduce formal technique for comparing algorithmic efficiency.

Better algorithm?

- For each number $A[i]$, look at $A[i+1]$, and return true the first time you see a match. If you run out of items, return false.

Clock Time

Lecture 11, CS61B, Fall 2025

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Intuitive Runtime Characterizations

- **Clock Time**
 - Exact Operation Counting
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- Big O

Intuitive Comparison: dup1 and dup2

Our goal is to somehow **characterize the runtimes** of the functions below.

- Intuitively, why is dup2 better?

```
public static boolean dup1(int[] A) {  
    for (int i = 0; i < A.length; i += 1) {  
        for (int j = i + 1; j < A.length; j += 1) {  
            if (A[i] == A[j]) {  
                return true;  
            }  
        }  
    }  
    return false;  
}
```

dup1

```
public static boolean dup2(int[] A) {  
    for (int i = 0; i < A.length - 1; i += 1) {  
        if (A[i] == A[i + 1]) {  
            return true;  
        }  
    }  
    return false;  
}
```

dup2

Intuitive Comparison: dup1 and dup2

Our goal is to somehow **characterize the runtimes** of the functions below.

- Dup2 is better because it does less work (takes less time) to arrive at the same result.

```
public static boolean dup1(int[] A) {  
    for (int i = 0; i < A.length; i += 1) {  
        for (int j = i + 1; j < A.length; j += 1) {  
            if (A[i] == A[j]) {  
                return true;  
            }  
        }  
    }  
    return false;  
}
```

dup1

```
public static boolean dup2(int[] A) {  
    for (int i = 0; i < A.length - 1; i += 1) {  
        if (A[i] == A[i + 1]) {  
            return true;  
        }  
    }  
    return false;  
}
```

dup2

How Do I Runtime Characterization?

Our goal is to somehow **characterize the runtimes** of the functions below.

- Characterization should be **simple** and **mathematically rigorous**.
- Characterization should **demonstrate superiority** of dup2 over dup1.

```
public static boolean dup1(int[] A) {  
    for (int i = 0; i < A.length; i += 1) {  
        for (int j = i + 1; j < A.length; j += 1) {  
            if (A[i] == A[j]) {  
                return true;  
            }  
        }  
    }  
    return false;  
}
```

dup1

```
public static boolean dup2(int[] A) {  
    for (int i = 0; i < A.length - 1; i += 1) {  
        if (A[i] == A[i + 1]) {  
            return true;  
        }  
    }  
    return false;  
}
```

dup2

Technique 1: Measure execution time in seconds using a client program.

- Tools:
 - Physical stopwatch.
 - Unix has a built in `time` command that measures execution time.
 - Call `System.currentTimeMillis()` before and after code runs.

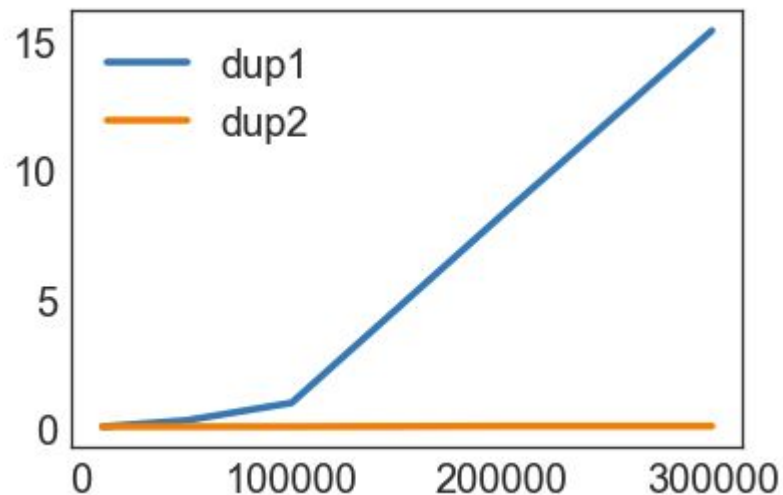
```
public static void main(String[] args) {  
    int N = Integer.parseInt(args[0]);  
    int[] A = makeArray(N);  
    dup1(A);  
}
```



Time Measurements for dup1 and dup2

N	dup1	dup2
10000	0.08	0.08
50000	0.32	0.08
100000	1.00	0.08
200000	8.26	0.1
400000	15.4	0.1

Time to complete (in seconds)

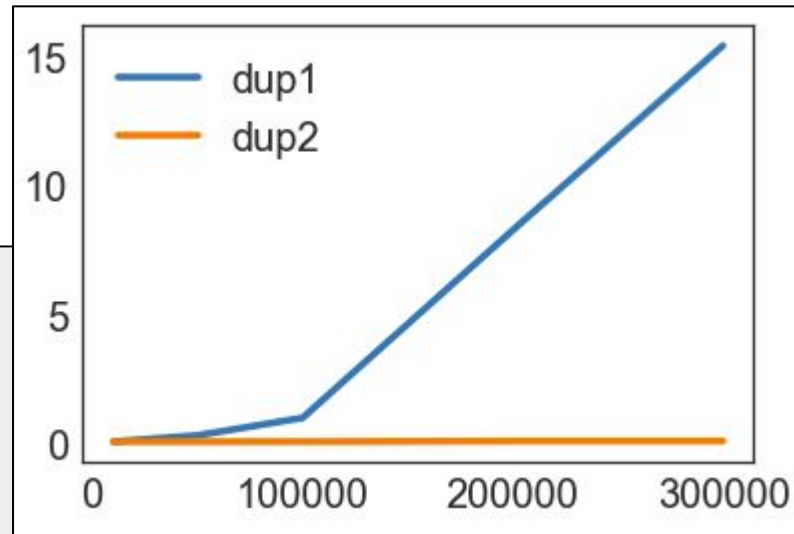


Techniques for Measuring Computational Cost

Technique 1: Measure execution time in seconds using a client program.

- Good: Easy to measure, meaning is obvious.
- Bad: May require large amounts of computation time. Result varies with machine, compiler, input data, etc.

```
public static void main(String[] args)
{
    int N = Integer.parseInt(args[0]);
    int[] A = makeArray(N);
    dup1(A);
}
```



Exact Operation Counting

Lecture 11, CS61B, Fall 2025

Goal: Measuring Code Efficiency

Intuitive Runtime Characterizations

- Clock Time
- **Exact Operation Counting**
- Exact Count Exercise

Asymptotic Analysis

- Why Scaling Matters
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Asymptotic Notation

- Big Theta (a.k.a. Order of Growth)
- Big O

Technique 2A: Count possible operations for an array of size $N = 10,000$.

- Good: Machine independent. Input dependence captured in model.
- Bad: Tedious to compute. Array size was arbitrary. Doesn't tell you actual time.

```
for (int i = 0; i < A.length; i += 1) {  
    for (int j = i+1; j < A.length; j += 1) {  
        if (A[i] == A[j]) {  
            return true;  
        }  
    }  
}  
return false;
```

operation	count, N=10000
i = 0	1
j = i + 1	1 to 10000
less than (<)	2 to 50,015,001
increment (+=1)	0 to 50,005,000
equals (==)	1 to 49,995,000
array accesses	2 to 99,990,000

The counts are tricky to compute. Work not shown. 

Technique 2B: Count possible operations in terms of input array size N.

- Good: Machine independent. Input dependence captured in model. Tells you how algorithm **scales**.
- Bad: Even more tedious to compute. Doesn't tell you actual time.

```
for (int i = 0; i < A.length; i += 1) {  
    for (int j = i+1; j<A.length; j += 1) {  
        if (A[i] == A[j]) {  
            return true;  
        }  
    }  
}  
return false;
```

operation	symbolic count	count, N=10000
i = 0	1	1
j = i + 1	1 to N	1 to 10000
less than (<)	2 to $(N^2+3N+2)/2$	2 to 50,015,001
increment (+=1)	0 to $(N^2+N)/2$	0 to 50,005,000
equals (==)	1 to $(N^2-N)/2$	1 to 49,995,000
array accesses	2 to N^2-N	2 to 99,990,000

Exact Count Exercise

Lecture 11, CS61B, Fall 2025

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Intuitive Runtime Characterizations

- Clock Time
- Exact Operation Counting
- **Exact Count Exercise**

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- Big O

Your turn: Try to come up with rough estimates for the symbolic and exact counts for at least one of the operations.

- Tip: Don't worry about being off by one. Just try to predict the rough magnitudes of each.

```
for (int i = 0; i < A.length - 1; i += 1){  
    if (A[i] == A[i + 1]) {  
        return true;  
    }  
}  
return false;
```

operation	sym. count	count, N=10000
i = 0	1	1
less than (<)		
increment (+=1)		
equals (==)		
array accesses		

Techniques for Measuring Computational Cost [dup2]

Your turn: Try to come up with rough estimates for the symbolic and exact counts for at least one of the operations.

```
for (int i = 0; i < A.length - 1; i += 1) {  
    if (A[i] == A[i + 1]) {  
        return true;  
    }  
}  
return false;
```

Especially observant folks may notice we didn't count everything, e.g. “- 1” and “+ 1” operations. We'll see why this omission is not a problem very shortly.

operation	symbolic count	count, N=10000
i = 0	1	1
less than (<)	1 to N	0 to 10000
increment (+=1)	0 to N - 1	0 to 9999
equals (==)	1 to N - 1	1 to 9999
array accesses	2 to 2N - 2	2 to 19998

If you did this exercise but were off by one, that's fine. The exact numbers aren't that important.

Why Scaling Matters

Lecture 11, CS61B, Fall 2025

Goal: Measuring Code Efficiency

Intuitive Runtime Characterizations

- Clock Time
- Exact Operation Counting
- Exact Count Exercise

Asymptotic Analysis

- **Why Scaling Matters**
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Comparing Algorithms

Which algorithm is better? dup2. Why?

- Fewer operations to do the same work [e.g. 50,015,001 vs. 10000 operations].
- Better answer: Algorithm **scales better** in the worst case. $(N^2+3N+2)/2$ vs. N .
- Even better answer: Parabolas (N^2) grow faster than lines (N).

operation	symbolic count	count, N=10000
i = 0	1	1
j = i + 1	1 to N	1 to 10000
less than (<)	2 to $(N^2+3N+2)/2$	2 to 50,015,001
increment (+=1)	0 to $(N^2+N)/2$	0 to 50,005,000
equals (==)	1 to $(N^2-N)/2$	1 to 49,995,000
array accesses	2 to N^2-N	2 to 99,990,000

dup1

operation	symbolic count	count, N=10000
i = 0	1	1
less than (<)	0 to N	0 to 10000
increment (+=1)	0 to N - 1	0 to 9999
equals (==)	1 to N - 1	1 to 9999
array accesses	2 to 2N - 2	2 to 19998

dup2

Which algorithm is better? Why?

operation	symbolic count	count, N=10000
i = 0	1	1
j = i + 1	1 to N	1 to 10000
less than (<)	2 to $(N^2+3N+2)/2$	2 to 50,015,001
increment (+=1)	0 to $(N^2+N)/2$	0 to 50,005,000
equals (==)	1 to $(N^2-N)/2$	1 to 49,995,000
array accesses	2 to N^2-N	2 to 99,990,000

dup1

operation	symbolic count	count, N=10000
i = 0	1	1
less than (<)	0 to N	0 to 10000
increment (+=1)	0 to N - 1	0 to 9999
equals (==)	1 to N - 1	1 to 9999
array accesses	2 to 2N - 2	2 to 19998

dup2

In most cases, we care only about asymptotic behavior, i.e. what happens for very large N .

- Simulation of billions of interacting particles.
- Social network with billions of users.
- Logging of billions of transactions.
- Encoding of billions of bytes of video data.

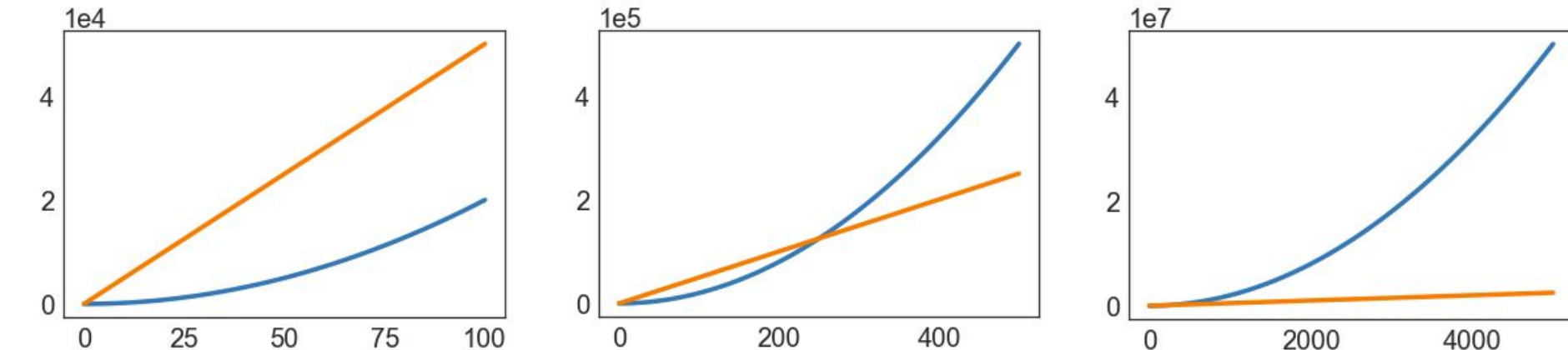
Algorithms which scale well (e.g. look like lines) have better asymptotic runtime behavior than algorithms that scale relatively poorly (e.g. look like parabolas).

Parabolas vs. Lines

Suppose we have two algorithms that zerpify a collection of N items.

- zerp1 takes $2N^2$ operations.
- zerp2 takes $500N$ operations.

For small N , zerp1 might be faster, but as dataset size grows, the parabolic algorithm is going to fall farther and farther behind (in time it takes to complete).



Scaling Across Many Domains

We'll informally refer to the “shape” of a runtime function as its **order of growth** (will formalize soon).

- Effect is dramatic! Often determines whether a problem can be solved at all.

	n	$n \log_2 n$	n^2	n^3	1.5^n	2^n	$n!$
$n = 10$	< 1 sec	< 1 sec	< 1 sec	< 1 sec	< 1 sec	< 1 sec	4 sec
$n = 30$	< 1 sec	< 1 sec	< 1 sec	< 1 sec	< 1 sec	18 min	10^{25} years
$n = 50$	< 1 sec	< 1 sec	< 1 sec	< 1 sec	11 min	36 years	very long
$n = 100$	< 1 sec	< 1 sec	< 1 sec	1 sec	12,892 years	10^{17} years	very long
$n = 1,000$	< 1 sec	< 1 sec	1 sec	18 min	very long	very long	very long
$n = 10,000$	< 1 sec	< 1 sec	2 min	12 days	very long	very long	very long
$n = 100,000$	< 1 sec	2 sec	3 hours	32 years	very long	very long	very long
$n = 1,000,000$	1 sec	20 sec	12 days	31,710 years	very long	very long	very long

Table 2.1 The running times (rounded up) of different algorithms on inputs of increasing size, for a processor performing a million high-level instructions per second. In cases where the running time exceeds 10^{25} years, we simply record the algorithm as taking a very long time.

(from Algorithm Design: Tardos, Kleinberg)

Our goal is to somehow **characterize the runtimes** of the functions below.

 Characterization should be **simple** and **mathematically rigorous**.

 Characterization should **demonstrate superiority** of dup2 over dup1.

operation	symbolic count
$i = 0$	1
$j = i + 1$	1 to N
less than ($<$)	2 to $(N^2 + 3N + 2)/2$
increment ($+=1$)	0 to $(N^2 + N)/2$
equals ($==$)	1 to $(N^2 - N)/2$
array accesses	2 to $N^2 - N$

dup1: parabolic, a.k.a. quadratic

operation	symbolic count
$i = 0$	1
less than ($<$)	0 to N
increment ($+=1$)	0 to N - 1
equals ($==$)	1 to N - 1
array accesses	2 to $2N - 2$

dup2: linear

Computing Worst Case Order of Growth (Tedious Approach)

Lecture 11, CS61B, Fall 2025

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Asymptotic Analysis

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- **Computing Worst Case Order of Growth (Tedious Approach)**
- Computing Worst Case Order of Growth (Simplified Approach)

Asymptotic Notation

- Big Theta (a.k.a. Order of Growth)
- Big O

Our goal is to somehow **characterize the runtimes** of the functions below.

- Characterization should be **simple** and **mathematically rigorous**.

operation	count
$i = 0$	1
$j = i + 1$	1 to N
less than ($<$)	2 to $(N^2 + 3N + 2)/2$
increment ($+=1$)	0 to $(N^2 + N)/2$
equals ($==$)	1 to $(N^2 - N)/2$
array accesses	2 to $N^2 - N$

operation	count
$i = 0$	1
less than ($<$)	0 to N
increment ($+=1$)	0 to $N - 1$
equals ($==$)	1 to $N - 1$
array accesses	2 to $2N - 2$

Let's be more careful about what we mean when we say the left function is “like” a parabola, and the right function is “like” a line.

Intuitive Simplification 1: Consider Only the Worst Case

Simplification 1: Consider only the worst case.

```
for (int i = 0; i < A.length; i += 1) {  
    for (int j = i+1; j < A.length; j += 1) {  
        if (A[i] == A[j]) {  
            return true;  
        }  
    }  
}  
return false;
```

operation	count
i = 0	1
j = i + 1	1 to N
less than (<)	2 to $(N^2+3N+2)/2$
increment (+=1)	0 to $(N^2+N)/2$
equals (==)	1 to $(N^2-N)/2$
array accesses	2 to N^2-N

Intuitive Simplification 1: Consider Only the Worst Case

Simplification 1: Consider only the worst case.

- **Justification:** When comparing algorithms, we often care only about the worst case [but we will see exceptions in this course].

```
for (int i = 0; i < A.length; i += 1) {  
    for (int j = i+1; j < A.length; j += 1) {  
        if (A[i] == A[j]) {  
            return true;  
        }  
    }  
}  
return false;
```

We're effectively focusing on the case where there are no duplicates, because this is where there is a performance difference.

operation	worst case count
$i = 0$	1
$j = i + 1$	N
less than ($<$)	$(N^2 + 3N + 2)/2$
increment ($+=1$)	$(N^2 + N)/2$
equals ($==$)	$(N^2 - N)/2$
array accesses	$N^2 - N$

Consider the algorithm below. What do you expect will be the **order of growth** of the runtime for the algorithm?

- A. N [linear]
- B. N^2 [quadratic]
- C. N^3 [cubic]
- D. N^6 [sextic]



operation	count
less than (<)	$100N^2 + 3N$
greater than (>)	$2N^3 + 1$
and (&&)	5,000

In other words, if we plotted total runtime vs. N , what shape would we expect?

Intuitive Order of Growth Identification

Consider the algorithm below. What do you expect will be the **order of growth** of the runtime for the algorithm?

A. N^3 [cubic]

operation	count
less than (<)	$100N^2 + 3N$
greater than (>)	$2N^3 + 1$
and (&&)	5,000

Argument:

- Suppose < takes α nanoseconds, > takes β nanoseconds, and && takes γ nanoseconds.
- Total time is $\alpha(100N^2 + 3N) + \beta(2N^3 + 1) + 5000\gamma$ nanoseconds.
- For very large N , the $2\beta N^3$ term is much larger than the others **regardless of the values of α , β , and γ .**

Extremely important point.
Make sure you understand it!

Intuitive Simplification 2: Restrict Attention to One Operation

Simplification 2: Pick some representative operation to act as a proxy for the overall runtime.

- Good choice: **increment**.  There are other good choices.
- Bad choice: assignment of $j = i + 1$.

We call our choice the “**cost model**”.

```
for (int i = 0; i < A.length; i += 1) {  
    for (int j = i+1; j < A.length; j += 1) {  
        if (A[i] == A[j]) {  
            return true;  
        }  
    }  
}  
return false;
```

cost model = **increment**



operation	worst case count
$i = 0$	1
$j = i + 1$	N
less than (<)	$(N^2 + 3N + 2)/2$
increment ($+=1$)	$(N^2 + N)/2$
equals (==)	$(N^2 - N)/2$
array accesses	$N^2 - N$

Intuitive Simplification 3: Eliminate low order terms.

Simplification 3: Ignore lower order terms.

```
for (int i = 0; i < A.length; i += 1) {  
    for (int j = i+1; j < A.length; j += 1) {  
        if (A[i] == A[j]) {  
            return true;  
        }  
    }  
}  
return false;
```

operation	worst case
increment ($+=1$)	$(N^2 + N)/2$

Intuitive Simplification 4: Eliminate multiplicative constants.

Simplification 4: Ignore multiplicative constants.

- Why? It has no real meaning. We already threw away information when we chose a single proxy operation.

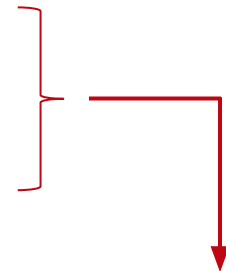
```
for (int i = 0; i < A.length; i += 1) {  
    for (int j = i+1; j < A.length; j += 1) {  
        if (A[i] == A[j]) {  
            return true;  
        }  
    }  
}  
return false;
```

operation	worst case
increment (+=1)	$N^2/2$

Simplification Summary

Simplifications:

1. Only consider the worst case.
2. Pick a representative operation (a.k.a. the cost model).
3. Ignore lower order terms.
4. Ignore multiplicative constants.



These three simplifications are OK because we only care about the “**order of growth**” of the runtime.

operation	count
$i = 0$	1
$j = i + 1$	1 to N
less than ($<$)	2 to $(N^2 + 3N + 2)/2$
increment ($+=1$)	0 to $(N^2 + N)/2$
equals ($==$)	1 to $(N^2 - N)/2$
array accesses	2 to $N^2 - N$



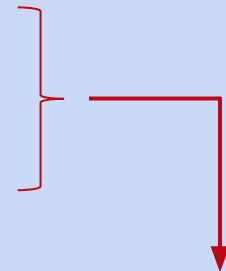
operation	worst case o.o.g.
increment ($+=1$)	N^2

Worst case order of growth of runtime: N^2

Simplification Summary: Repeating the Process for dup2

Simplifications:

1. Only consider the worst case.
2. Pick a representative operation (a.k.a. the cost model).
3. Ignore lower order terms.
4. Ignore multiplicative constants.



These three simplifications are OK because we only care about the “**order of growth**” of the runtime.

operation	count
$i = 0$	1
less than ($<$)	0 to N
increment ($+=1$)	0 to $N - 1$
equals ($==$)	1 to $N - 1$
array accesses	2 to $2N - 2$



operation	worst case o.o.g.

Worst case order of growth of runtime:

Simplifications:

1. Only consider the worst case.
2. Pick a representative operation (a.k.a. the cost model).
3. Ignore lower order terms.
4. Ignore multiplicative constants.

This simplification is OK because we specifically only care about worst case.

These three simplifications are OK because we only care about the “**order of growth**” of the runtime.

operation	count
$i = 0$	1
less than ($<$)	0 to N
increment ($+=1$)	0 to $N - 1$
equals ($==$)	1 to $N - 1$
array accesses	2 to $2N - 2$

operation	worst case o.o.g.
array accesses	N

Worst case order of growth of runtime: N

Any of the bottom four operations are good choices.

Summary of Our (Painful) Analysis Process

Our process:

- Construct a table of exact counts of all possible operations.
- Convert table into a worst case order of growth using 4 simplifications.

operation	count
$i = 0$	1
$j = i + 1$	1 to N
less than ($<$)	2 to $(N^2 + 3N + 2)/2$
increment ($+=1$)	0 to $(N^2 + N)/2$
equals ($==$)	1 to $(N^2 - N)/2$
array accesses	2 to $N^2 - N$



operation	worst case o.o.g.
increment ($+=1$)	N^2

Worst case order of growth of runtime: N^2

By using our simplifications from the outset, we can avoid building the table at all!

Computing Worst Case Order of Growth (Simplified Approach)

Lecture 11, CS61B, Fall 2025

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- Clock Time
- Exact Operation Counting
- Exact Count Exercise

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- Why Scaling Matters
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Asymptotic Notation

- Big Theta (a.k.a. Order of Growth)
- Big O

Rather than building the entire table, we can instead:

- Choose a representative operation to count (a.k.a. cost model).
- Figure out the order of growth for the count of the representative operation by either:
 - Making an exact count, then discarding the unnecessary pieces.
 - Using intuition and inspection to determine order of growth (only possible with lots of practice).

Let's redo our analysis of dup1 with this new process.

- This time, we'll show all our work.

Analysis of Nested For Loops (Based on Exact Count)

Find the order of growth of the worst case runtime of dup1.

N = 6

0		==	==	==	==	==
1			==	==	==	==
2				==	==	==
3					==	==
4						==
5						
	0	1	2	3	4	5

i

j

```
int N = A.length;
for (int i = 0; i < N; i += 1)
    for (int j = i + 1; j < N; j += 1)
        if (A[i] == A[j])
            return true;
return false;
```

Worst case number of == operations:

$$C = 1 + 2 + 3 + \dots + (N - 3) + (N - 2) + (N - 1)$$

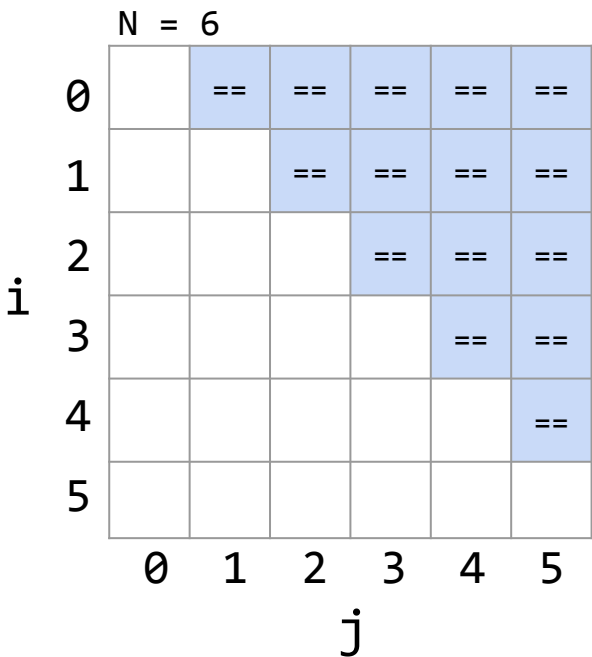
$$C = (N - 1) + (N - 2) + (N - 3) + \dots + 3 + 2 + 1$$

$$2C = \underbrace{N + N + \dots + N}_{N-1 \text{ of these}} = N(N - 1)$$

$$\therefore C = N(N - 1)/2$$

Analysis of Nested For Loops (Based on Exact Count)

Find the order of growth of the worst case runtime of dup1.



```
int N = A.length;
for (int i = 0; i < N; i += 1)
    for (int j = i + 1; j < N; j += 1)
        if (A[i] == A[j])
            return true;
return false;
```

Worst case number of == operations:

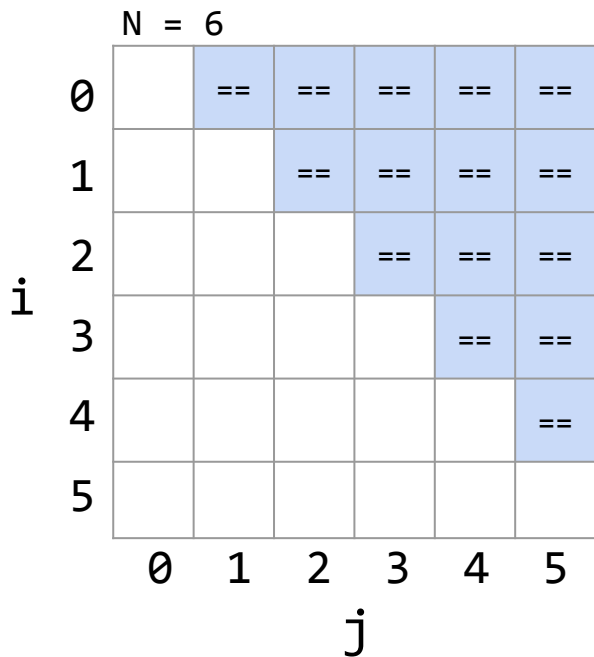
$$C = 1 + 2 + 3 + \dots + (N - 3) + (N - 2) + (N - 1) = N(N-1)/2$$

operation	worst case o.o.g.
==	N ²

Worst case order of growth of runtime: N²

Analysis of Nested For Loops (Simpler Geometric Argument)

Find the order of growth of the worst case runtime of dup1.



```
int N = A.length;
for (int i = 0; i < N; i += 1)
    for (int j = i + 1; j < N; j += 1)
        if (A[i] == A[j])
            return true;
return false;
```

Worst case number of == operations:

- Given by area of right triangle of side length N-1.
- Order of growth of area is N^2 .

operation	worst case o.o.g.
==	N^2

Worst case order of growth of runtime: N^2

Big Theta (a.k.a. Order of Growth)

Lecture 11, CS61B, Fall 2025

Goal: Measuring Code Efficiency

Intuitive Runtime Characterizations

- Clock Time
- Exact Operation Counting
- Exact Count Exercise

Asymptotic Analysis

- Why Scaling Matters
- Computing Worst Case Order of Growth (Tedious Approach)
- Computing Worst Case Order of Growth (Simplified Approach)

Asymptotic Notation

- **Big Theta (a.k.a. Order of Growth)**
- Big O

Formalizing Order of Growth

Given a function $Q(N)$, we can apply our last two simplifications (ignore low orders terms and multiplicative constants) to yield the order of growth of $Q(N)$.

- Example: $Q(N) = 3N^3 + N^2$
- Order of growth: N^3

Let's finish out this lecture by moving to a more formal notation called Big-Theta.

- The math might seem daunting at first.
- ... but the idea is exactly the same! Using “Big-Theta” instead of “order of growth” does not change the way we analyze code at all.

Order of Growth Exercise

function	order of growth
$N^3 + 3N^4$	
$1/N + N^3$	
$1/N + 5$	
$N^N + N$	
$40 \sin(N) + 4N^2$	

Order of Growth Exercise

Consider the functions below.

- Informally, what is the “shape” of each function for very large N ?
- In other words, what is the order of growth of each function?

function	order of growth
$N^3 + 3N^4$	N^4
$1/N + N^3$	N^3
$1/N + 5$	1
$N^e + N$	N^e
$40 \sin(N) + 4N^2$	N^2

Suppose we have a function $R(N)$ with order of growth $f(N)$.

- In “Big-Theta” notation we write this as $R(N) \in \Theta(f(N))$.
- Examples:
 - $N^3 + 3N^4 \in \Theta(N^4)$
 - $1/N + N^3 \in \Theta(N^3)$
 - $1/N + 5 \in \Theta(1)$
 - $Ne^N + N \in \Theta(Ne^N)$
 - $40 \sin(N) + 4N^2 \in \Theta(N^2)$

function $R(N)$	order of growth
$N^3 + 3N^4$	N^4
$1/N + N^3$	N^3
$1/N + 5$	1
$Ne^N + N$	Ne^N
$40 \sin(N) + 4N^2$	N^2

$$R(N) \in \Theta(f(N))$$

means there exist positive constants k_1 and k_2 such that:

$$k_1 \cdot f(N) \leq R(N) \leq k_2 \cdot f(N)$$

for all values of N greater than some N_0 .

 i.e. very large N

Example: $40 \sin(N) + 4N^2 \in \Theta(N^2)$

- $R(N) = 40 \sin(N) + 4N^2$
- $f(N) = N^2$
- $k_1 = 3$
- $k_2 = 5$

Suppose $R(N) = (4N^2 + 3N \ln(N))/2$.

- Find a simple $f(N)$ and corresponding k_1 and k_2 .

$$R(N) \in \Theta(f(N))$$

means there exist positive constants k_1 and k_2 such that:

$$k_1 \cdot f(N) \leq R(N) \leq k_2 \cdot f(N)$$

for all values of N greater than some N_0 .

← i.e. very large N

Suppose $R(N) = (4N^2 + 3N \ln(N))/2$.

- $f(N) = N^2$
- $k_1 = 1$
- $k_2 = 3$

$$R(N) \in \Theta(f(N))$$

means there exist positive constants k_1 and k_2 such that:

$$k_1 \cdot f(N) \leq R(N) \leq k_2 \cdot f(N)$$

for all values of N greater than some N_0 .

← i.e. very large N

Big-Theta and Runtime Analysis

Using Big-Theta doesn't change anything about runtime analysis (no need to find k_1 or k_2 or anything like that).

- The only difference is that we use the Θ symbol anywhere we would have said "order of growth".

operation	worst case count
$i = 0$	1
$j = i + 1$	$\Theta(N)$
less than ($<$)	$\Theta(N^2)$
increment ($+=1$)	$\Theta(N^2)$
equals ($==$)	$\Theta(N^2)$
array accesses	$\Theta(N^2)$



operation	worst case count
increment ($+=1$)	$\Theta(N^2)$

Worst case runtime: $\Theta(N^2)$

Big O (time permitting)

Lecture 11, CS61B, Fall 2025

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- Exact Count Exercise

Asymptotic Analysis

- Why Scaling Matters
- Computing Worst Case Order of Growth (Tedious Approach)
- Computing Worst Case Order of Growth (Simplified Approach)

Asymptotic Notation

- Big Theta (a.k.a. Order of Growth)
- **Big O (time permitting)**

We used Big Theta to describe the order of growth of a function.

function $R(N)$	order of growth
$N^3 + 3N^4$	$\Theta(N^4)$
$1/N + N^3$	$\Theta(N^3)$
$1/N + 5$	$\Theta(1)$
$N^N + N$	$\Theta(N^N)$
$40 \sin(N) + 4N^2$	$\Theta(N^2)$

We also used Big Theta to describe the rate of growth of the runtime of a piece of code.

Whereas Big Theta can informally be thought of as something like “equals”, Big O can be thought of as “less than or equal”.

Example, the following are all true:

- $N^3 + 3N^4 \in \Theta(N^4)$
- $N^3 + 3N^4 \in O(N^4)$
- $N^3 + 3N^4 \in O(N^6)$
- $N^3 + 3N^4 \in O(N!)$
- $N^3 + 3N^4 \in O(N^{N!})$

$$R(N) \in \Theta(f(N))$$

means there exist positive constants k_1 and k_2 such that:

$$k_1 \cdot f(N) \leq R(N) \leq k_2 \cdot f(N)$$

for all values of N greater than some N_0 .

 i.e. very large N

Example: $40 \sin(N) + 4N^2 \in \Theta(N^2)$

- $R(N) = 40 \sin(N) + 4N^2$
- $f(N) = N^2$
- $k_1 = 3$
- $k_2 = 5$

$$R(N) \in O(f(N))$$

means there exists positive constants k_2 such that:

$$R(N) \leq k_2 \cdot f(N)$$

for all values of N greater than some N_0 .

 i.e. very large N

Example: $40 \sin(N) + 4N^2 \in O(N^4)$

- $R(N) = 40 \sin(N) + 4N^2$
- $f(N) = N^4$
- $k_2 = 1$

Big Theta vs. Big O

We will see why big O is practically useful in the upcoming Disjoint Sets lecture.

	Informal meaning:	Family	Family Members
Big Theta $\Theta(f(N))$	Order of growth is $f(N)$.	$\Theta(N^2)$	$N^2/2$ $2N^2$ $N^2 + 38N + N$
Big O $O(f(N))$	Order of growth is less than or equal to $f(N)$.	$O(N^2)$	$N^2/2$ $2N^2$ $\lg(N)$

Given a code snippet, we can express its runtime as a function $R(N)$, where N is some property of the input of the function (often the size of the input).

Rather than finding $R(N)$ exactly, we instead usually only care about the order of growth of $R(N)$.

One approach (not universal):

- Choose a representative operation, and let $C(N)$ be the count of how many times that operation occurs as a function of N .
- Determine order of growth $f(N)$ for $C(N)$, i.e. $C(N) \in \Theta(f(N))$
 - Often (but not always) we consider the worst case count.
- If operation takes constant time, then $R(N) \in \Theta(f(N))$
- Can use O as an alternative for Θ . O is used for upper bounds.

TSP problem solution, title slide:

http://support.sas.com/documentation/cdl/en/ornoaug/65289/HTML/default/viewer.htm#ornoaug_optnet_examples07.htm#ornoaug.optnet.map002g

Table of runtimes for various orders of growth: Kleinberg & Tardos, Algorithm Design.

Cost model terminology: Algorithms 4th edition by Robert Sedgewick and Kevin Wayne

Attendance link: <https://www.yellkey.com/help>

To-do:

- Project 2, due today.
- Study for midterm.
 - Covers lectures 1 - 9.
 - Complete HWs and projects.
 - Complete discussion worksheets week 1 - 3.
 - Go through old exam problems.

